4.2 BJT Static Analysis

4.2.1 Calculate Carrier Densities

4.2.2 Calculate Current Densities

4.2.3 Circuit Parameters

4.2.4 Narrow Base Approximations

Pierret, Chapter 11, page 389-404

BJT – Ideal Static Analysis I

Goal:

- Calculate BJT **currents** and **transistor parameters** α_0, I_{CB0}, β_0

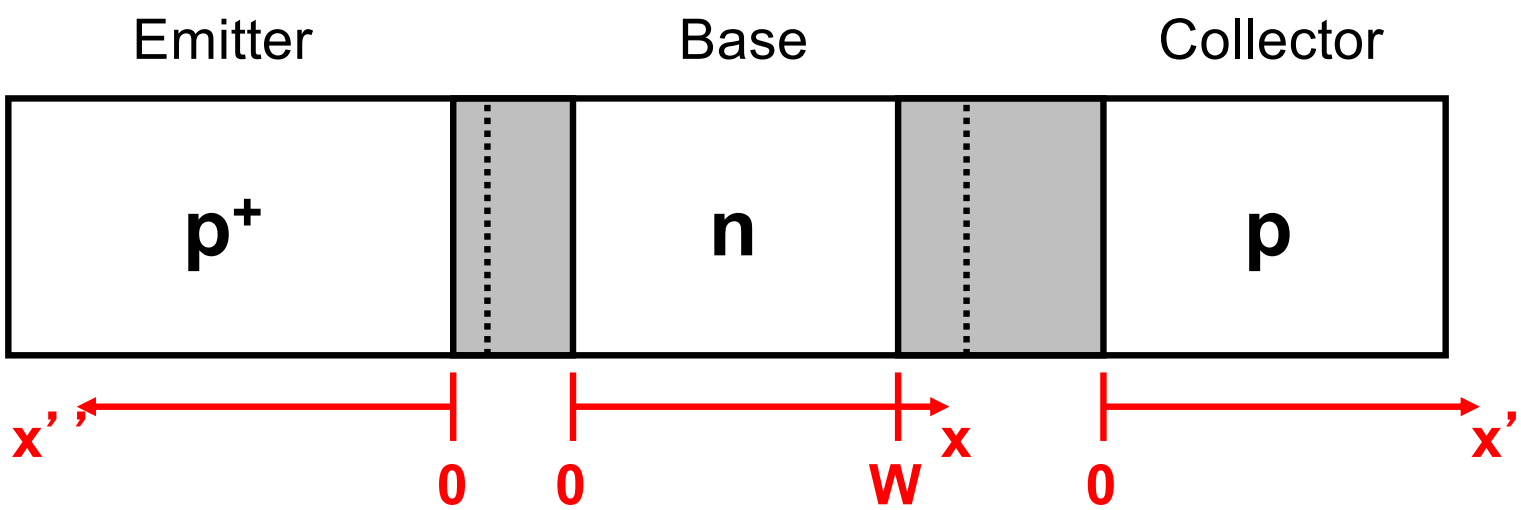
Assumptions (similar to pn-junction analysis):

- 1-D model with homogeneous doping, i.e. $N_A(x)$, $N_D(x)$ = const.; as a result, the junctions are treated as step junctions
- Low-level injection
- No R-G currents in the depletion regions
- Non-degenerate doping, i.e. Boltzmann approximation is valid
- Depletion layer width << base width
- No electric field outside the depletion regions

Solution Approach (similar to pn-junction analysis):

1. Calculate carrier densities in quasineutral regions
2. Calculate current densities at the edges of the depletion regions

BJT – Ideal Static Analysis III
– The Model –



$N_E = N_{A,E}$	$N_B = N_{D,B}$	$N_C = N_{A,C}$
$D_E = D_n$	$D_B = D_p$	$D_C = D_n$
$L_E = L_n$	$L_B = L_p$	$L_C = L_n$
$n_{E0} = n_{p0}$	$p_{B0} = p_{n0}$	$n_{C0} = n_{p0}$
$\tau_E = \tau_n$	$\tau_B = \tau_p$	$\tau_C = \tau_n$

4.2.1 Calculate Carrier Densities

Approach: Solve **steady-state continuity equations** in the quasi-neutral regions (no electric field, no carrier generation) of emitter, base, and collector

$$\frac{\partial n_p(x,t)}{\partial t} = +n_p\mu_n \frac{\partial \mathcal{E}}{\partial x} + \mu_n \mathcal{E} \frac{\partial n_p}{\partial x} + \mathbf{D_n} \frac{\partial^2 \mathbf{n_p}}{\partial \mathbf{x}^2} + G_n - \frac{\mathbf{n_p} - \mathbf{n_{p0}}}{\tau_n}$$

$$\underbrace{\frac{\partial p_n(x,t)}{\partial t}}_{\substack{=0 \\ \text{steady state}}} = \underbrace{-p_n\mu_p \frac{\partial \mathcal{E}}{\partial x} - \mu_p \mathcal{E} \frac{\partial p_n}{\partial x}}_{\substack{=0 \\ \text{no electric field}}} + \underbrace{\mathbf{D_p} \frac{\partial^2 \mathbf{p_n}}{\partial \mathbf{x}^2}}_{\textbf{Diffusion}} + \underbrace{G_p}_{\substack{=0 \\ \text{no G}}} - \underbrace{\frac{\mathbf{p_n} - \mathbf{p_{n0}}}{\tau_p}}_{\textbf{Thermal R-G}}$$

Carrier Density – Emitter I

- Continuity equation for emitter:

$$0 = D_E \frac{d^2 \Delta n_E(x'')}{dx''^2} - \frac{\Delta n_E(x'')}{\tau_E}$$

or

$$0 = D_E \frac{d^2 n_E(x'')}{dx''^2} - \frac{n_E(x'') - n_{E0}}{\tau_E}$$

- General solution:

$$\Delta n_E(x'') = n_E(x'') - n_{E0} = A_1 e^{-x''/L_E} + A_2 e^{+x''/L_E}$$

with $L_E = \sqrt{D_E \tau_E}$

Carrier Density – Emitter II

- Boundary Conditions (see pn-junction):

$$\Delta n_E(0) = n_{E0}\left[e^{qV_{EB}/kT} - 1\right]$$

$$\Delta n_E(\infty) = 0$$

- Emitter Solution:

$$\boxed{\Delta n_E(x'') = n_{E0}\left[e^{qV_{EB}/kT} - 1\right]e^{-x''/L_E}}$$

Carrier Density – Collector

- Calculation similar to emitter!
- Minority carrier distribution in quasi-neutral collector region:

$$\boxed{\Delta n_C(x') = n_{C0}\left[e^{qV_{CB}/kT} - 1\right]e^{-x'/L_C}}$$

Carrier Density – Base I

- Continuity equation:

$$0 = D_B \frac{d^2 \Delta p_B(x)}{dx^2} - \frac{\Delta p_B(x)}{\tau_B}$$

- Boundary conditions:

$$\Delta p_B(0) = p_{B0}\left[e^{qV_{EB}/kT} - 1\right]$$

$$\Delta p_B(W) = p_{B0}\left[e^{qV_{CB}/kT} - 1\right]$$

- General solution:

$$\Delta p_B(x) = A_1 e^{-x/L_B} + A_2 e^{+x/L_B}$$

Carrier Density – Base II

- Minority carrier distribution in the quasi-neutral base region:

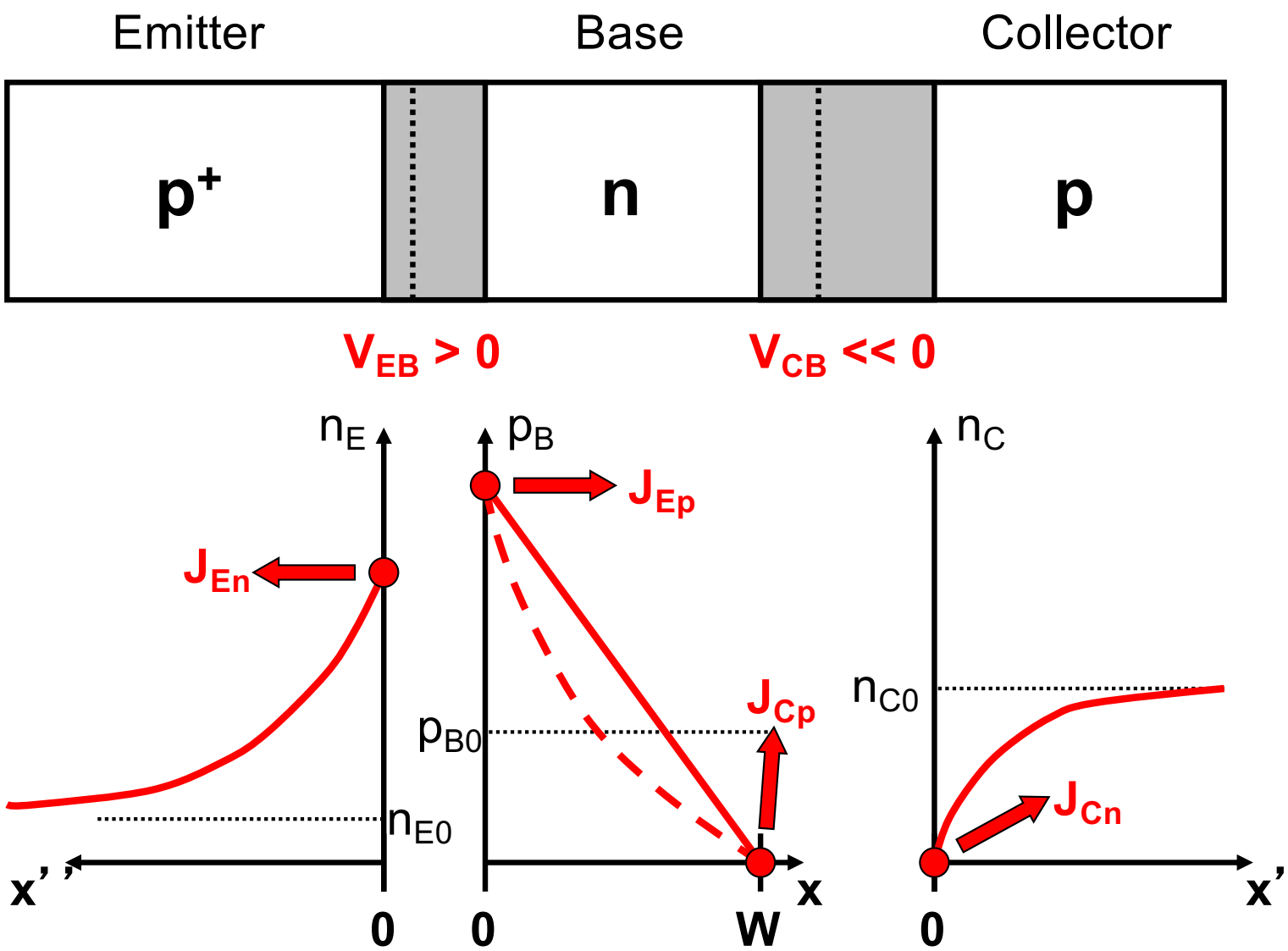
$$\Delta p_B(x) = \Delta p_B(0)\left[\frac{e^{(W-x)/L_B} - e^{-(W-x)/L_B}}{e^{W/L_B} - e^{-W/L_B}}\right] + \Delta p_B(W)\left[\frac{e^{x/L_B} - e^{-x/L_B}}{e^{W/L_B} - e^{-W/L_B}}\right]$$

$$\sinh(\xi) = \frac{e^{\xi} - e^{-\xi}}{2}$$

$$\cosh(\xi) = \frac{e^{\xi} + e^{-\xi}}{2}$$

$$\Delta p_B(x) = \Delta p_B(0)\left[\frac{\sinh\left[(W-x)/L_B\right]}{\sinh\left[W/L_B\right]}\right] + \Delta p_B(W)\left[\frac{\sinh\left[x/L_B\right]}{\sinh\left[W/L_B\right]}\right]$$

Carrier Densities



4.2.2 Calculate Current Densities

- **Approach:** Calculate current densities at the edges of the depletion regions from the minority carrier distributions in the quasi-neutral regions, thus considering only carrier diffusion

$$J_p = q\,\mu_p\,p\,\varepsilon - q\,D_p\,\nabla p$$

$$J_n = \underbrace{q\,\mu_n\,n\,\varepsilon}_{\substack{=0 \\ \text{no E-field}}} + q\,D_n\,\nabla n$$

Current Densities J_{En} & J_{Cn}

From current density equation for electrons, we obtain for the minority carrier (electron) currents in emitter and collector at the edge of the depletion layer (x' = x" = 0):

$$J_{Cn}(x' = 0) = +q\,D_C \left.\frac{d\Delta n_C}{dx'}\right|_{x'=0} = -q\,\frac{D_C}{L_C}\,n_{C0}\,(\underbrace{e^{qV_{CB}/kT}}_{\approx 0} - 1)$$

$$J_{En}(x'' = 0) = -q\,D_E \left.\frac{d\Delta n_E}{dx''}\right|_{x''=0} = +q\,\frac{D_E}{L_E}\,n_{E0}\,(\underbrace{e^{qV_{EB}/kT}}_{>>1} - 1)$$

"–" sign for emitter because of "negative" coordinate system

Current Densities J_{Ep} and J_{Cp}

Calculated from the minority carrier current density in the base region (at the edges of the E-B and B-C depletion regions, x = 0 and x = W), assuming no R-G effects in the depletion region, i.e. no current changes across the depletion regions:

$$J_{Ep}\left(x'' = 0\right) = J_{Bp}(x = 0) = -q\,D_B \left.\frac{d\Delta p_B}{dx}\right|_{x=0}$$

$$= q\,\frac{D_B}{L_B}\,p_{B0}\left[\frac{\cosh\left(W/L_B\right)}{\sinh\left(W/L_B\right)}\left(e^{qV_{EB}/kT} - 1\right) - \frac{1}{\sinh\left(W/L_B\right)}\left(e^{qV_{CB}/kT} - 1\right)\right]$$

$$J_{Cp}\left(x' = 0\right) = J_{Bp}(x = W) = -q\,D_B \left.\frac{d\Delta p_B}{dx}\right|_{x=W}$$

$$= q\,\frac{D_B}{L_B}\,p_{B0}\left[\frac{1}{\sinh\left(W/L_B\right)}\left(e^{qV_{EB}/kT} - 1\right) - \frac{\cosh\left(W/L_B\right)}{\sinh\left(W/L_B\right)}\left(e^{qV_{CB}/kT} - 1\right)\right]$$

BJT Currents

$$I_E = I_{Ep} + I_{En} = A\left(J_{Ep} + J_{En}\right)$$

$$I_C = I_{Cp} + I_{Cn} = A\left(J_{Cp} + J_{Cn}\right)$$

$$I_B = I_E - I_C$$

with BJT cross-sectional area A and current densities J_{Ep}, J_{En}, J_{Cp}, and J_{Cn} from previous pages!

4.2.3 Circuit Parameters I

- **Emitter Efficiency:**

$$\gamma = \frac{I_{Ep}}{I_{Ep} + I_{En}} = \frac{1}{1 + \left(\frac{D_E}{D_B}\frac{L_B}{L_E}\frac{N_B}{N_E}\right)\frac{\sinh(W/L_B)}{\cosh(W/L_B)}}$$

with $p_{B0} = n_i^2/N_B$ and $n_{E0} = n_i^2/N_E$

- **Base Transport Factor:**

$$\alpha_T = \frac{I_{Cp}}{I_{Ep}} = \frac{1}{\cosh\left(W/L_B\right)}$$

Circuit Parameters II

- **Common Base Current Gain:**

$$\alpha_0 = \gamma\,\alpha_T = \frac{1}{\cosh(W/L_B) + \left(\frac{D_E}{D_B}\frac{L_B}{L_E}\frac{N_B}{N_E}\right)\sinh(W/L_B)}$$

- **Common Emitter Current Gain:**

$$\beta_0 = \frac{\alpha_0}{1-\alpha_0} = \frac{1}{\cosh(W/L_B) + \left(\frac{D_E}{D_B}\frac{L_B}{L_E}\frac{N_B}{N_E}\right)\sinh(W/L_B) - 1}$$

4.2.4 Small Base W « L_B Approximation

- **Minority carrier distribution in the base:**

$$\Delta p_B(x) = \Delta p_B(0)\left[1 - \frac{x}{W}\right] + \underbrace{\Delta p_B(W)\left[\frac{x}{W}\right]}_{\substack{= -p_{B0}(x/W) \text{ for large} \\ \text{reverse bias of B–C}}}$$

with sinh x ≈ x and cosh x ≈ 1 + 1/2 x^2 for small x

- **Emitter Efficiency:**

$$\gamma = \frac{1}{1 + \left(\frac{D_E}{D_B}\frac{W}{L_E}\frac{N_B}{N_E}\right)}$$

γ becomes close to 1 if N_E » N_B

Small Base W « L_B Approximation (cont.)

- **Base Transport Factor:**

$$\alpha_T = \frac{1}{1 + \frac{1}{2}\left(\frac{W}{L_B}\right)^2}$$

α_T becomes close to 1 if L_B » W

- **Current Gains** (neglecting higher order terms):

$$\alpha_0 = \frac{1}{1 + \left(\frac{D_E}{D_B}\frac{W}{L_E}\frac{N_B}{N_E}\right) + \frac{1}{2}\left(\frac{W}{L_B}\right)^2}$$

$$\beta_0 = \frac{1}{\left(\frac{D_E}{D_B}\frac{W}{L_E}\frac{N_B}{N_E}\right) + \frac{1}{2}\left(\frac{W}{L_B}\right)^2}$$

Small Base W « L_B Approximation (cont.)

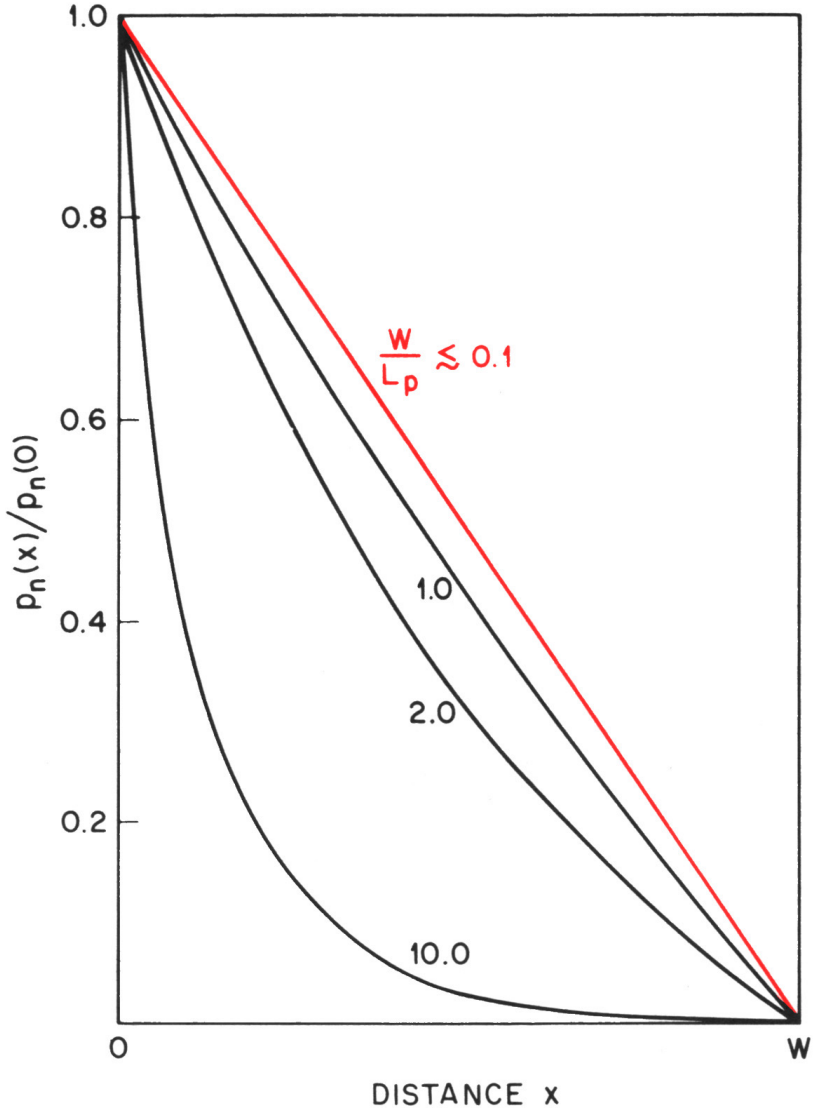
- **The currents** (active mode: $V_{EB} > 0$, $V_{CB} < 0$):

$$I_E = \overbrace{\left(qA\frac{D_B}{W}p_{B0}\right)e^{qV_{EB}/kT}}^{=I_{Ep}} + \overbrace{\left(qA\frac{D_E}{L_E}n_{E0}\right)\left(e^{qV_{EB}/kT}-1\right)}^{=I_{En}}$$

$$I_C = \underbrace{\left(qA\frac{D_B}{W}p_{B0}\right)e^{qV_{EB}/kT}}_{=I_{Cp}} + \underbrace{\left(qA\frac{D_C}{L_C}n_{C0}\right)}_{=I_{Cn}}$$

$$I_B = I_E - I_C = \left(qA\frac{D_E}{L_E}n_{E0}\right)\left(e^{qV_{EB}/kT}-1\right) - \left(qA\frac{D_C}{L_C}n_{C0}\right)$$

Minority carrier densities

Minority Carrier Density in Base

- In case of a small base compared to the minority carrier diffusion length in the base ($W \ll L_B$), the minority carrier concentration in active mode decreases linearly across the base from the forward-biased E-B to the reverse-biased B-C junction
- A linear decrease in $p_B(x)$ means $I_{Ep} = I_{Cp}$, i.e. very few minority carriers recombine while diffusing across the base; this is also expressed in a base transport factor close to 1